



Mixture

It is composed of different substances that do not have constant composition.

Mixtures can be separated into different substances by physical means without changing the identities of the components. Ex: Air, coffee, cement

Homogenous Mixture

This is the type of mixture that is uniform all throughout.

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Heterogeneous Mixture

This is the type of mixture in which the components are not uniform.

Colloids

These are mixtures in which the particles do not settle at the bottom upon standing.

Suspension

These are mixtures in which the particles settle at the bottom upon standing.



Mixture Separation

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- Decantation or Floatation** – it is the process of separating large solid particles from liquids by allowing the liquid to flow while the large solid particles to settle.
- Filtration** – it is the process of separating small solid particles from liquid by passing the mixture through a porous medium.
- Distillation** – it is the process of separating the liquid by boiling the mixture to vaporize the liquid and then cooling the vapor to condense it.
 - Fractional distillation** – it is the process of separating liquid mixtures whose component substances have narrow differences in boiling point
 - Steam distillation** – it is the process of separating volatile oils from plants

- Mechanical separation** – it is the process involving the use of tools such as forceps, sieves, etc.

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- Centrifugation** – it is the process that speeds up the settling of the precipitates using a centrifuge which is a motor-driven apparatus

- Chromatography** – it is the process of using the difference in degree to which the substances are absorbed on the surface of an inert substance

- Solvent extraction** – it is the process of separating the mixture to its component substances by making use of the difference in solubility of the substances

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- Amalgamation** – it is the process of extracting gold from its ore

- Cyanidation** – it is the process of extracting gold from its ore by using cyanide

Measurements

Measurement is important in engineering science as it allows properties to be defined in a quantitative value.



Macroscopic properties are properties that can be measured directly while **microscopic properties** are properties that can be measured using an indirect method. Some typical measuring devices used in chemistry are flasks, pipets, and graduated cylinders.

International System of Units - SI

K	▶ kelvin (temperature)
m	▶ meter (distance)
A	▶ ampere (electric current)
s	▶ second (time)
mol	▶ mole (amount of substance)
kg	▶ kilogram (mass)
cd	▶ candela (intensity of light)

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The SI system or the International System of Units) is the revised metric system of units proposed by the General Conference of Weights and Measures in 1960. It has 7 base units and all other units can be derived from these base units.

7 basic SI units

Quantity	Unit	Symbol
Mass	kilogram	Kg
Length	meter	m
Time	second	s
Temperature	Kelvin	K
Electric Current	Ampere	A
Amount of Substance	mole	mol
Luminous Intensity	candela	Cd

The SI system is based on the decimal system. Listed in the table below are prefixes used to extend the range of SI units.

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Prefix	Symbol	Factor	Notation
Giga	G	1,000,000,000	10^9
Mega	M	1,000,000	10^6
Kilo	K	1,000	10^3
Deci	d	1/10	10^{-1}
Centi	C	1/100	10^{-2}
Milli	M	1/1,000	10^{-3}
Micro	μ	1/1,000,000	10^{-6}
Nano	n	1/1,000,000,000	10^{-9}
Pico	p	1/1,000,000,000,000	10^{-12}

The Meter



For the meter, which is the standard unit or metric unit of length, it is usually defined as 10^{-7} of the distance from the equator to the North Pole along a meridian passing Paris, France. It is also defined as the distance between two parallel marks in a specially made platinum-iridium bar. Listed below are useful lengths/distances expressed in meters:

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	Length/Distance
Radius of a hydrogen atom	5.3×10^{-11}
1 Angstrom	10^{-10}
1 inch	0.0254
1 mile	1609
Radius of the Earth	6.38×10^6
Distance between Earth and Moon	3.84×10^8
Radius of the Sun	6.97×10^8
1 Astronomical Unit (Distance between the Sun and the Earth)	1.5×10^{11}
1 Lightyear	9.46×10^{15}

The Kilogram

For the kilogram, which is the standard unit of mass, it is defined as the mass of a specially made cylinder of platinum-iridium.



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